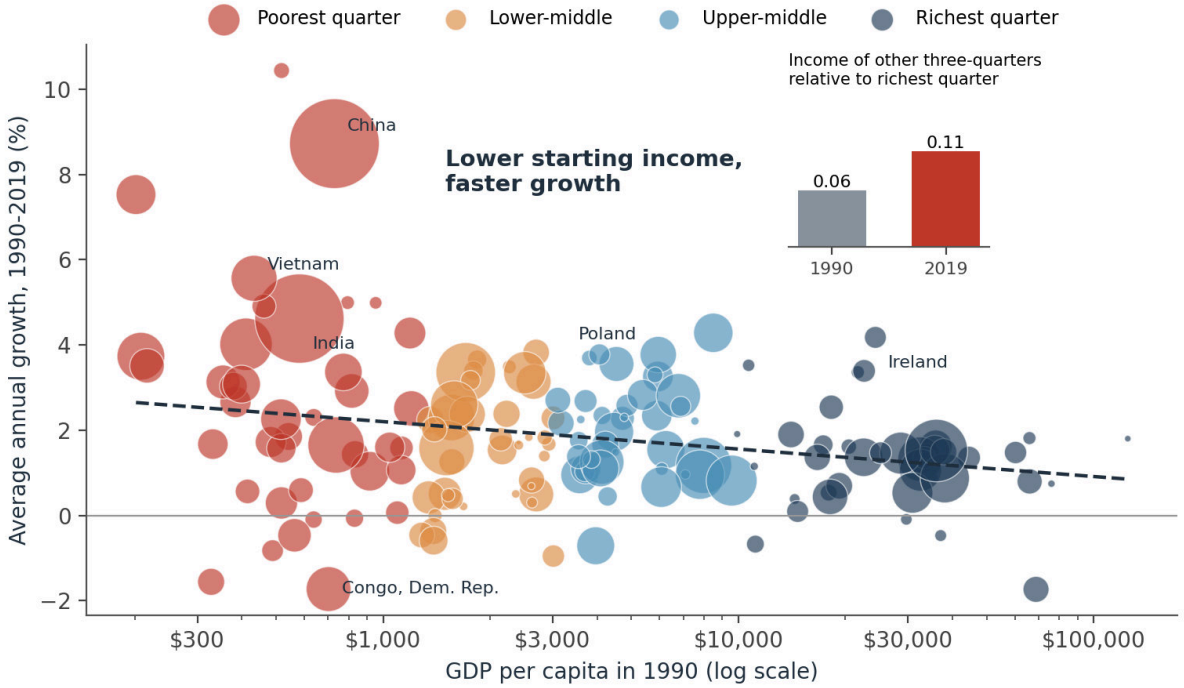
Side A: the proposition is TRUE

Proposition: "Poorer countries are catching up with richer ones" (GDP per capita, 1990-2019)

Countries that started poorer have grown faster

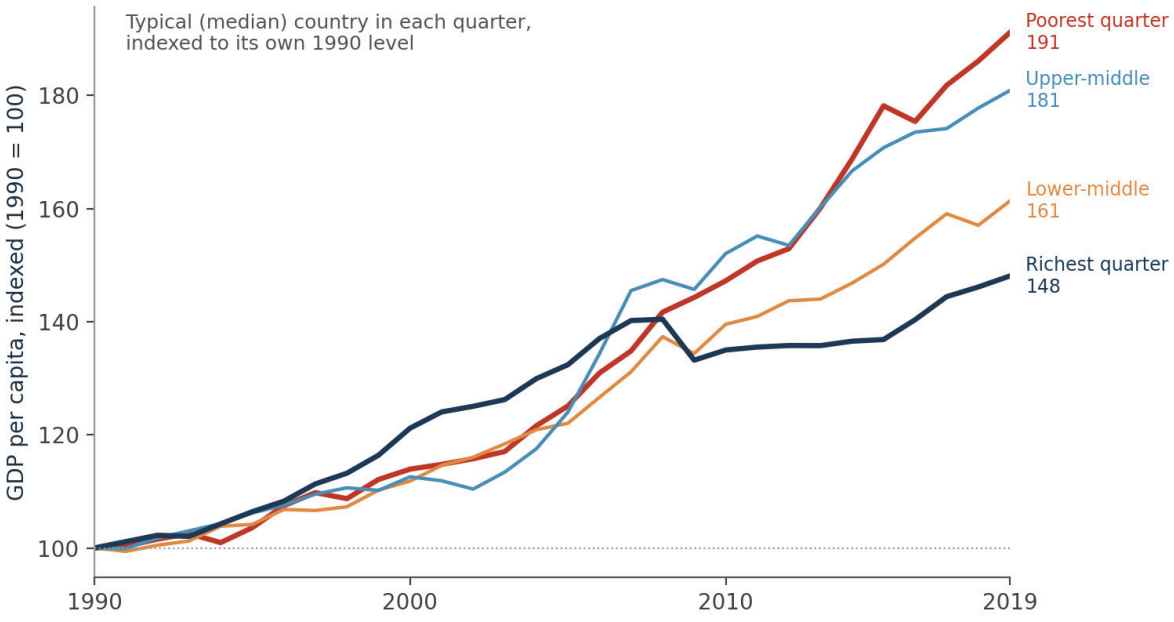
Annual GDP per capita growth against starting income, 1990-2019. Bubble size = population.



Source: World Bank, GDP per capita (constant 2010 US$). 162 countries with data in 1990 and 2019; quarters = countries ranked by 1990 GDP per capita.

The poorest countries have grown fastest since 1990

Median change in GDP per capita relative to each country's own 1990 level.



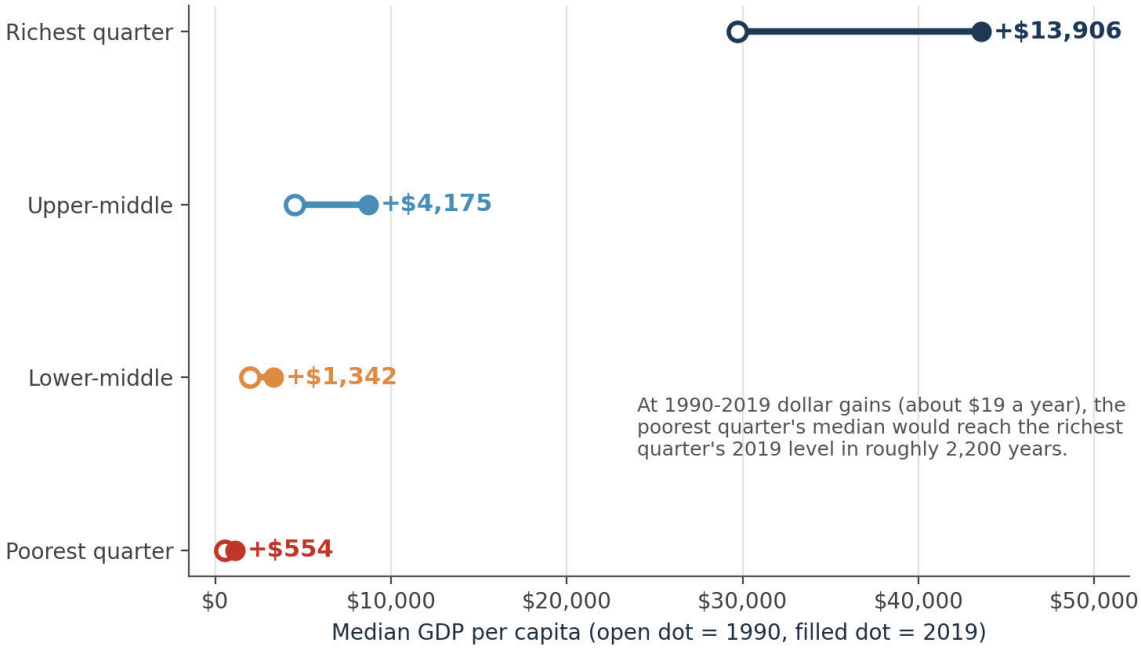
Source: World Bank, GDP per capita (constant 2010 US$). 162 countries with data in 1990 and 2019; quarters = countries ranked by 1990 GDP per capita.

Side B: the proposition is FALSE

Same proposition, opposing stance. Same data: 162 countries, World Bank GDP per capita.

Dollars added per person, 1990-2019

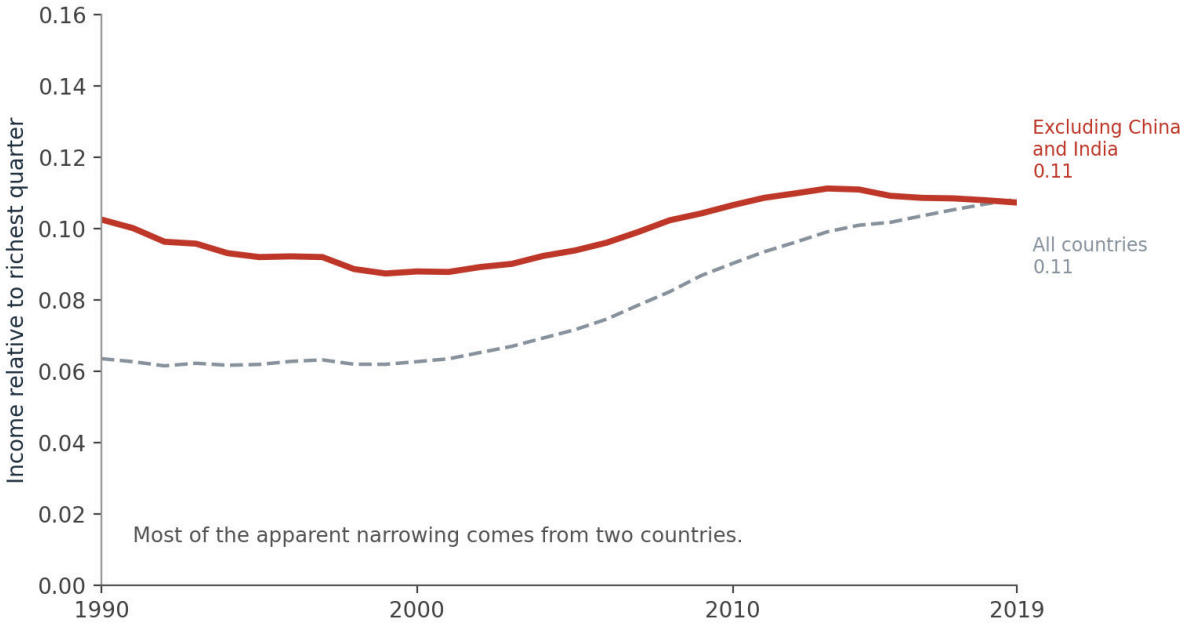
Median GDP per capita by starting-income quarter, in constant 2010 US dollars.



Source: World Bank, GDP per capita (constant 2010 US$). 162 countries with data in 1990 and 2019; quarters = countries ranked by 1990 GDP per capita.

Relative income of the other three-quarters, 1990-2019

Population-weighted GDP per capita of the lower three starting-income quarters, as a share of the richest quarter.



Source: World Bank, GDP per capita (constant 2010 US$). 162 countries with data in 1990 and 2019; quarters = countries ranked by 1990 GDP per capita.

PROJECT 2 CHECKPOINT: WRITE-UP

Proposition (same on both sides)
"Poorer countries are catching up with richer ones." The data is World Bank GDP per capita (constant 2010 US$) for the 162 countries that have values in both 1990 and 2019, plus population. I grouped countries into quarters by their 1990 income.

Which set I'm leaning toward
I'm arguing both sides from the same data, so the final will use one chart per side. Right now I prefer A1 (growth vs. starting income) for TRUE and B1 (dollars added) for FALSE. If there's room I'll fold B2 into B1 as an inset. A2 and B2 are backups.

Side A (TRUE): deceptive techniques
- A1 measures growth as a percent rate instead of dollars. A $500 gain is a big percentage for a poor country and a rounding error for a rich one, so this favors low starting points.
- A1 sizes bubbles by population, so China and India dominate the picture. The inset also uses population weights: the poorer three-quarters go from 0.06 to 0.11 of the richest quarter's income.
- A1 uses a log x-axis, which spreads out the poorest countries and squeezes the rich ones together.
- A1 fixes the window at 1990-2019. With a 1970 start the correlation between starting income and growth is -0.07 (it is -0.25 from 1990), and 1990-2000 alone slopes the other way. The trend line is drawn prominently even though the relationship is weak.
- A2 indexes every country to its own 1990 level (a calculated field) and plots the median per quarter. Indexing hides that the starting incomes differ by about 50x.
- A2 shows only the median country, so the large dollar gaps never appear. The title says "poorest countries" when it describes the median one.

Side B (FALSE): deceptive techniques
- B1 measures absolute dollars added instead of percent growth, which favors high starting points. The richest quarter's median GDP per capita rose about $13,900. The poorest quarter's rose about $550.
- B1 uses a linear axis, so the poorest group's two dots overlap and look static.
- B1 extrapolates the poorest quarter's dollar gain in a straight line to claim it would take roughly 2,200 years to match the richest quarter. Straight-line extrapolation over that span is a stretch, and I count it as deceptive.
- B2 drops China and India from the lower group, so the weighted ratio ends almost where it started (0.103 in 1990, 0.107 in 2019, versus 0.064 to 0.108 with everyone). The label says what was removed, but I chose which countries to drop. Dropping China alone removes about three-quarters of the narrowing, and India alone removes almost none, so the "two countries" framing leans on China.
- B2 runs its y-axis from 0 to 0.16, which flattens the line.

Across all four charts
- Titles and annotations are plain and neutral. The slant comes from what is measured, grouped, or left out.
- All four use the same 162-country panel, so the differences come from design choices and not from different data.
- Transformations: compound annual growth rate, binning into quarters, indexing to 1990, population-weighted ratios, and a ratio that excludes two countries.